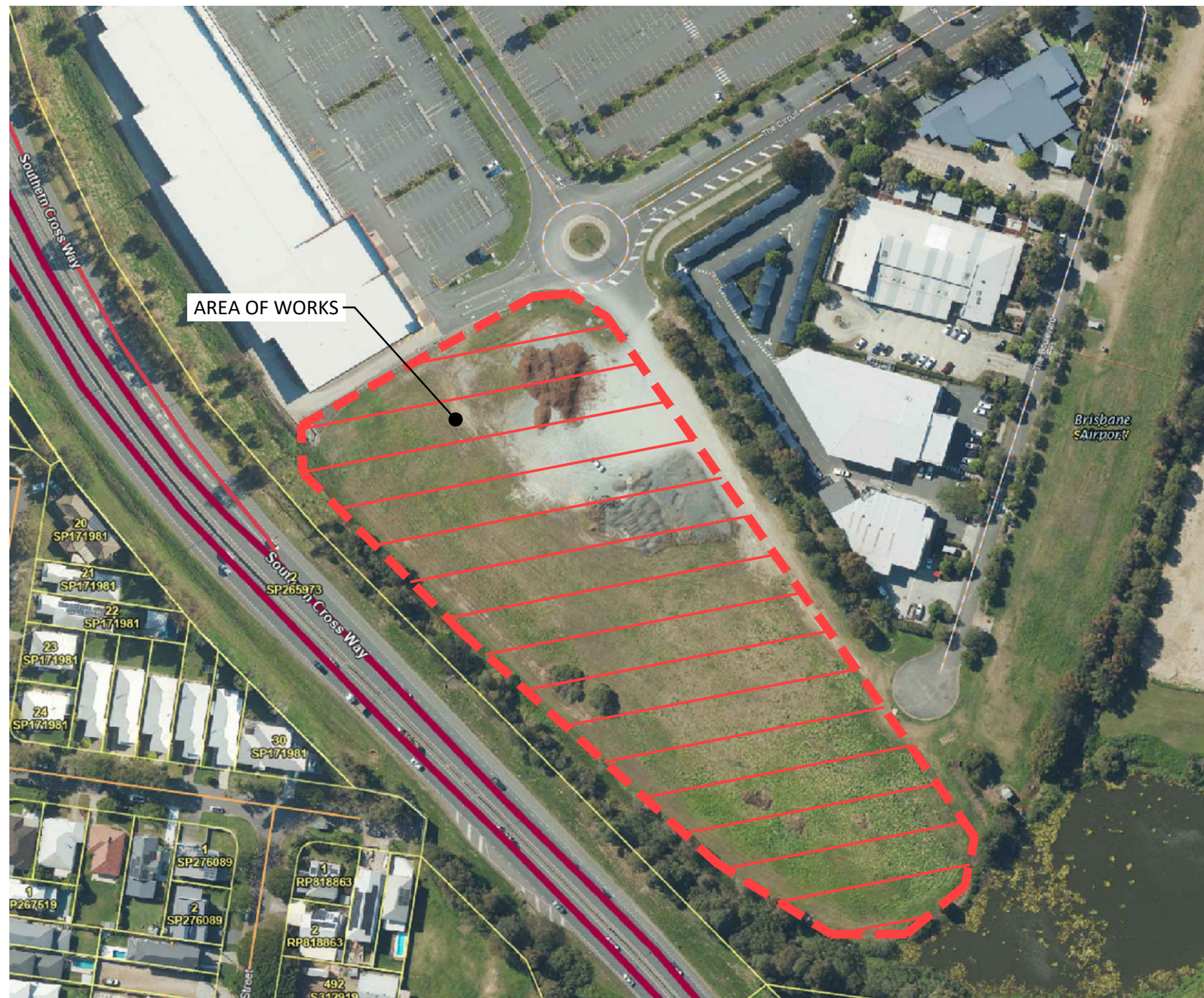


25-0437 - BNE PLAY

THE CIRCUIT, BRISBANE AIRPORT QLD 4008



PROJECT WET FIRE DETAILS	
N.C.C. VERSION	2022
FIRE HYDRANT STANDARD	AS2419.1-2021
PROJECT CLASSIFICATION	9B, 6
RISE IN STOREYS	1
TYPE OF CONSTRUCTION	C - LARGE ISOLATED BUILDING
NUMBER OF LEVELS	1
LARGEST FIRE COMPARTMENT AREA	7180m ²
SPRINKLER PROTECTION?	YES
FHR REQUIRED?	YES
TYPE OF HYDRANT SYSTEM	ATTACK
No. OF HYDRANTS REQUIRED TO OPERATE SIMULTANEOUSLY	2
MIN. ATTACK HYDRANT SYSTEM PERFORMANCE	2x 5 L/s @ 700 kPa (PUMPED) 2x 10 L/s @ 700 kPa (QFES BOOST)
PREPARED AS PER OMNII FIRE ENGINEERING BRIEF DATED 16/12/25	

FLOW & PRESSURE TESTING REPORT DATA	
FLOW (L/s)	95th PERCENTILE PRESSURE (kPa)
0	532
10	512
20	480
30	472
40 (FULL FLOW)	400

TAKEN FROM 95TH PERCENTILE FLOW & PRESSURE
TEST REPORT SAN26 PREPARED BY BAC ON 02/10/25.

FIRE DESIGN AS DOCUMENTED IN THESE DRAWINGS IS
BASED ON THE MORE CONSERVATIVE OF THE ABOVE
RESULTS TO AVOID PROBLEMS AT COMMISSIONING.

PHYSICAL PROTECTION OF PILLAR HYDRANTS

QFES DOES NOT SUPPORT THE USE OF HYDRANT SYSTEMS THAT ARE MOUNTED ON, PASS THROUGH, OR ARE FIXED TO EXTERNAL TILT-UP PANEL WALLS IN BUILDINGS. CONTRACTOR SHALL ALLOW TO ENSURE SUITABLE MOUNTING & FIXING POINTS ARE PROVIDED TO SUPPORT THE FIRE SYSTEM PIPEWORK. EXTERNAL HYDRANTS SHALL BE PROTECTED BY BOLLARDS.

REAL PROPERTY DESCRIPTION	
LOT 10 ON SP 238997	
SOIL CLASSIFICATION	
SITE CLASSIFICATION	CLASS P
SOIL PROPERTIES	CLASS M
Ys RANGE	30-40mm
BASED UPON GEOTECHNICAL REPORT 229444.00.R.001.REV0 PREPARED BY "DOUGLAS PARTNERS" DATED 11 MARCH 2025	

NOTE

ALLOW ALL COSTS TO FIX PIPEWORK
UNDERNEATH SLAB VIA STAINLESS STEEL
RODS TO SUIT UNSTABLE SOIL CONDITIONS

DEFINED FLOOD LEVEL THE DEFINED FLOOD LEVEL IS RL 1.900m AS PER LOCAL COUNCIL FLOOD LEVEL REPORT DATED 21/08/25
--

CAUTION

OTHER SERVICES EXIST ON-SITE. CONTRACTOR IS TO CONDUCT "DIAL BEFORE YOU DIG" AND SURVEY PRIOR TO COMMENCEMENT OF WORKS & MAKE ON-SITE ROUTE ADJUSTMENTS AS REQUIRED

SYMBOLS

	FLOW DIRECTION
	SERVICE DESCRIPTION
	SERVICE SIZE
	FLOW DIRECTION
	DIRECTION OF FLOW
	CONTINUATION MARKER
	CONNECT TO EXISTING
	DROPPER
	RISE
	TEE
	CAPPED PIPE
	FLOOR WASTE
	BUCKET TRAP
	OVERFLOW RELIEF GULLY
	TUNDISH
	REFLUX VALVE
	RAINWATER OUTLET
	PLANTER DRAIN OUTLET
	AIR ADMITTANCE VALVE
	COS C/W HEAVY DUTY TRAFFICABLE LID
	MANHOLE
	STORMWATER PIT
	GRATED TRENCH DRAIN
	FIXTURE OUTLET
	HOSE TAP
	WATER METER
	ISOLATION VALVE
	MONITORED VALVE
	CHECK VALVE
	DOUBLE CHECK VALVE
	DUAL CHECK VALVE ASSEMBLY
	REDUCED PRESSURE ZONE DEVICE
	PRESSURE REDUCTION VALVE
	TEMPERATURE LIMITING VALVE
	THERMOSTATIC MIXING VALVE
	HOT WATER UNIT
	INSTANTANEOUS HOT WATER UNIT
	PUMP
	EXPANSION CONTROL VALVE
	STRAINER
	VIBRATION JOINT
	UNION
	FLANGE
	FIRE HOSE REEL
	HYDRANT
	DUAL PILLAR HYDRANT
	BRIGADE BOOST
	BRIGADE SUCTION
	TANK SUCTION
	HIGH POINT
	EXPANSION AND/OR SWIVEL JOINT

LINETYPES		
PROPOSED	EXISTING	
	sd	SANITARY DRAINAGE
	tw	TRADE WATER
	vw	VENT
	sw	STORMWATER
	cw	COLD WATER
	ww	WARM WATER
	hw	HOT WATER
	rw	RAINWATER
	fs	FIRE SERVICE
	npcw	NOT POTABLE COLD WATER
	picw	PIPEWORK IN SLAB
		CONCRETE CUT
	c	CONDUIT (GENERAL)
		DECOMMISSION LINE
	cs	CIVIL SEWER
	csw	CIVIL STORMWATER
	cww	CIVIL WATER SERVICE
		FIRE WALL
		SMOKE WALL

ABBREVIATIONS

ACV	CELISUS	FSL	FINISHED SURFACE LEVEL	RHS	RECTANGULAR HOLLOW SECTION
AAV	AIR ADMITTANCE VALVE	FSR	FIRE SERVICE RISER	RLH	REDUCED LEVEL
ACH	AIR CONDITIONER	FDW	FIRE TEST	RPZD	REDUCED PRESSURE ZONE DEVICE
ABD	ABOVE FINISHED FLOOR LEVEL	FTU	FIXTURE UNIT	RFV	REFLEX VALVE
AHD	AUSTRALIAN HEIGHT DATUM	FWD	FLOOR WASTE GULLY	RW	RAINWATER
ACP	ACCESS PANEL	GA	GALVANISED STEEL	RWD	RAINWATER DROPPER
B	BASIN	GIT	GREASE INTERCEPTOR TRAP	RWO	RAINWATER OUTLET
BCWU	BOILING/CHILLED WATER UNIT	GTD	GRAATED TRENCH DRAIN	RWR	RAINWATER RISER
BT	BUCK TRAP	GTV	GREASE TRAP VENT PIPE	S	SEWER
BU	BULKING WATER	HDPE	HIGH DENSITY POLYETHYLENE	S/S	STAINLESS STEEL
BWU	BOILING WATER UNIT	HP	HIGH POINT	SD	SANITARY DRAINAGE
CB	CERAMIC PLATE	HP/E	HIGH POINT/EXPANSION JOINT	SH	STRUCTURAL FLOOR LEVEL
C/C	COMPLETE WITH	HT	HOSE TAP	SHR	SHOWER
CD	CONDENSATE DRAIN	HW	HOT WATER	SIC	SEWER INSPECTION CHAMBER
CHV	CHECK VALVE	HWD	HOT WATER DROPPER	SI	SWIVEL JOINT
CHWU	CHILLED WATER UNIT	HW	HOT WATER RISER	SK	SINK
CI	CAST IRON	HWU	HOT WATER UNIT	SL	SURFACE LEVEL
CLPL	CLAY PLUG	IC	INSPECTION CHAMBER	SMH	SEWER MANHOLE
CM	CORNER MACHINE	IV	INVERT LEVEL	SPR	SPREADER
CS	CLEAROUT	IO	INSPECTION OPENING	ST	STACK
CO	CLEAROUT TO SURFACE	IOS	INSPECTION OPENING TO SURFACE	STR	STRAINER
CSK	CLEANERS SINK	IU	ISOLATION VALVE	SV	SOLENOID VALVE
CTE	CONNECT TO EXISTING	JV	JUMP UP	SW	STORMWATER
COP	COPPER	KPS	KILOPASCALS	SWD	STORMWATER DROPPER
CW	COLD WATER	KW	KILOWATT	SWIP	STORMWATER INLET PIT
CWD	COLD WATER DROPPER	L	LITRE	SWMH	STORMWATER MANHOLE
CWR	COLD WATER RISER	L/h	LITRES PER HOUR	TD	TUNDISH
DIS	DISABLED	L/h	LITRES PER SECOND	TM	TEMPERATURE LIMITING VALVE
DCHV	DUAL CHECK VALVE	LAU	LAUNDRY TUB	TMV	THERMOSTATIC MIXING VALVE
DCV	DOUBLE CHECK VALVE	LU	LOADING UNITS	PRV	TEMPERATURE & PRESSURE RELIEF VALVE
DF	DRINKING FOUNTAIN	m	METRE(S)	TTD	TRAPPED TUNDISH
DISC	DISCONNECTOR GULLY	m/s	METRES PER SECOND	TTG	TEST GATE
DN	NOMINAL DIAMETER	MPE	MEDIUM DENSITY POLYETHYLENE	TW	TRADE WASTE
DP	DOWNPIPE	MJ	MEGAJOULE	TWEP	TRADE WASTE ELEVATED
DPH	DUAL PILLAR HYDRANT	mm	MILLIMETRE(S)	TWST	TRADE WASTE STACK
DS	DOUBLE SINK	mv	MONITORED WASTE	TWVP	TRADE WASTE VENT PIPE
DTU	DRAINAGE TURN-UP	NF	NOT POTABLE COLD WATER	TYP	TYPICAL
DW	DISHWASHER	NPCWD	NOT POTABLE COLD WATER DROPPER	UNO	UNUSUAL NOT OTHERWISE
EW	EXISTING	NPCWR	NOT POTABLE COLD WATER RISER	UPVC	UNPLASTICISED POLYVINYL CHLORIDE
ECV	EXPANSION CONTROL VALVE	NPWS	NOT POTABLE WATER SERVICE	UR	URINAL
EJ/SJ	EXPANSION & SWIVEL JOINT	NZS	NEW ZEALAND STANDARD	VB	VACUUM BREAKER
EXP	EXPANSION JOINT	O/F	OVERFLOW	VP	VENT PIPE
EPD	ELEVATED PIPE DROPPER	ORF	OVERFLOW RELIEF GULLY	WC	WATER CLOSET
FFL	FINISHED FLOOR LEVEL	P	PLANTER DRAIN	WM	WATER METER
FH	FIRE HYDRANT	PLD	PLANTER DRAIN OUTLET	WW	WARM WATER
FR	FIRE HOSE REEL	PLO	PLANTER LIMITING VALVE	WWD	WARM WATER DROPPER
FR	REFRIGERATOR	PRV	PRESSURE REDUCTION VALVE	WWR	WARM WATER RISER
FSD	FIRE SERVICE DROPPER	PVC	POLYVINYL CHLORIDE		
		RE	RELOCATE		

Sheet List Table	
Sheet Number	Sheet Title
000000	LEGEND, NOTES & LOCALITY PLAN
001101	GROUND FLOOR DRAINAGE LAYOUT 1
001102	GROUND FLOOR DRAINAGE LAYOUT 2
001103	ROOF DRAINAGE LAYOUT 1
001104	ROOF DRAINAGE LAYOUT 2
001201	GROUND FLOOR WATER LAYOUT 1
001202	GROUND FLOOR WATER LAYOUT 1
001203	GROUND FLOOR WATER INSETS - SHEET 1 OF 2
001204	GROUND FLOOR WATER INSETS - SHEET 2 OF 2

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	BW	D&C PRICING ISSUE	04/02/26	B
	BW	D&C PRICING ISSUE	18/12/25	A
	DRN	AMENDMENTS	DATE	REV

25-0437

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HYDRAULIC SERVICES
QBCC LICENCE No: 15126403

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y Ltd and the instructing party.

Drawn BW	Checked
Design Coordinated IS	Compliance Checked
Authorised For Issue BD	Construction Checked

SKYGATE PLAY
THE CIRCUIT, BRISBANE AIRPORT

HYDRAULIC SERVICES

LEGEND, NOTES & LOCALITY PLAN

	Brisbane Airport Corporation Pty Ltd 11 The Circuit Brisbane Airport QLD 4008 PO Box 61 Hamilton QLD 4007 Australia						Scale NTS
	Telephone: 61 7 3406 3000 Facsimile: 61 7 3406 3111 Email: info@bne.com.au www.brisbaneairport.com.au						Size A1
	ARN 54 076 870 650						Revision B
Drawing Number	Project No.	Originator	Volume	Level	Type	Discipline	Number
25021	-	ASH	-	XX	-	XX - DR - HY	- 000000

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25021	-	ASH	-	XX	-	XX - DR - HY	- 000000

D&C PRICING ISSUE

This drawing is diagrammatic & approximate only. The (sub-)contractor shall make all allowances necessary to price the project on a Design & Construct methodology. Where items are not specifically shown or included, but are required, the (sub-)contractor shall make all allowances for such items to ensure compliant system(s) are provided that satisfy the client's expectations & operational requirements.